OpenClaw Servo Control Tutorial

> Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on servo-related content.

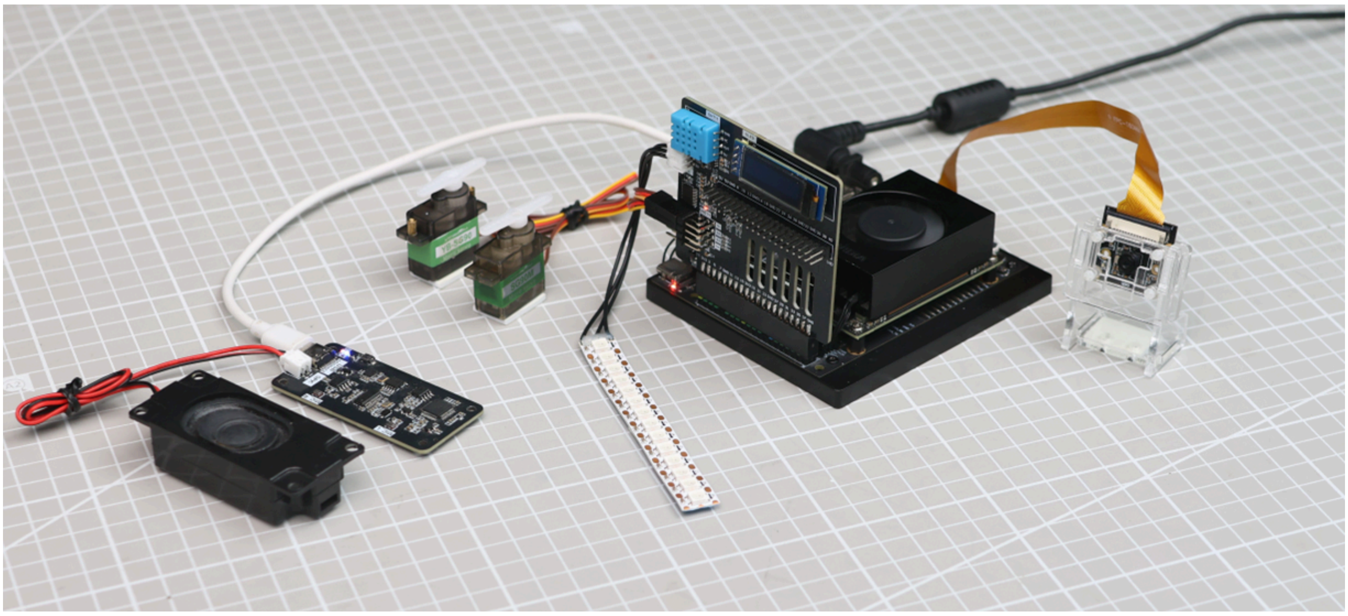
1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capabilities through servo peripherals. After completing this tutorial, readers can:

- Understand the basic approach to controlling servos with OpenClaw
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four servo cases: single servo control, servo dancing, yes/no question answering, table tennis simulation, pointer simulation

2. Hardware Preparation

- OpenClaw mainboard
- Servo `[1 piece / 4 pieces]`
- `[Optional] Microphone module`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Low-level driver code: `/home/jetson/hardware_hal_demo/`

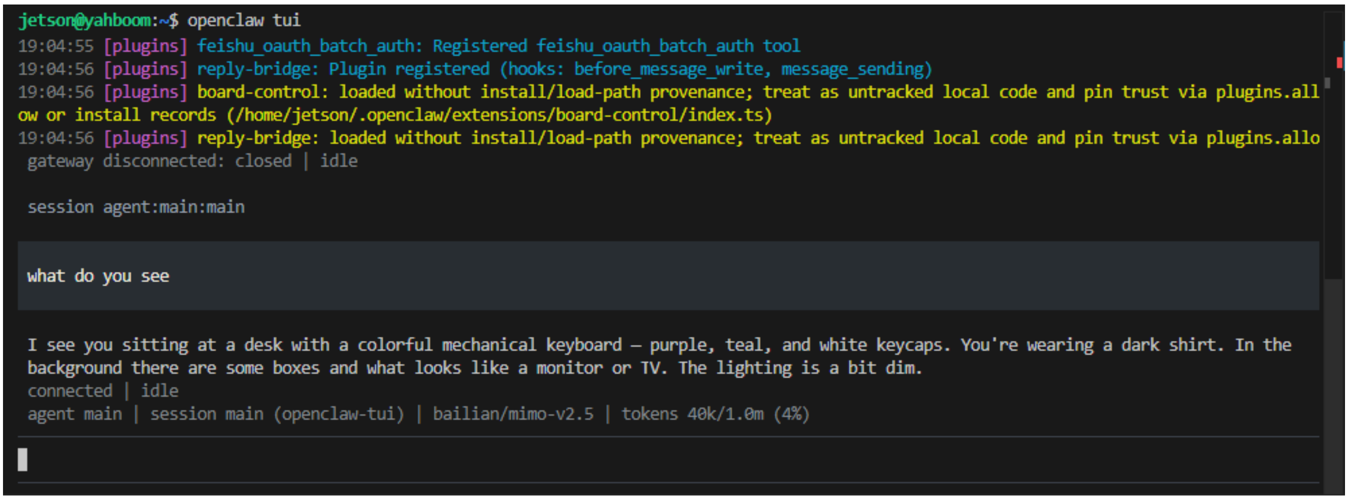
- Python driver library: `/home/jetson/Downloads/python/`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/jetson/voice_to_openclaw/src/text_main.py`

4. Test Servo

Before starting this tutorial, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After confirming OpenClaw can respond normally, enter servo-specific testing.

Terminal input to enter TUI:

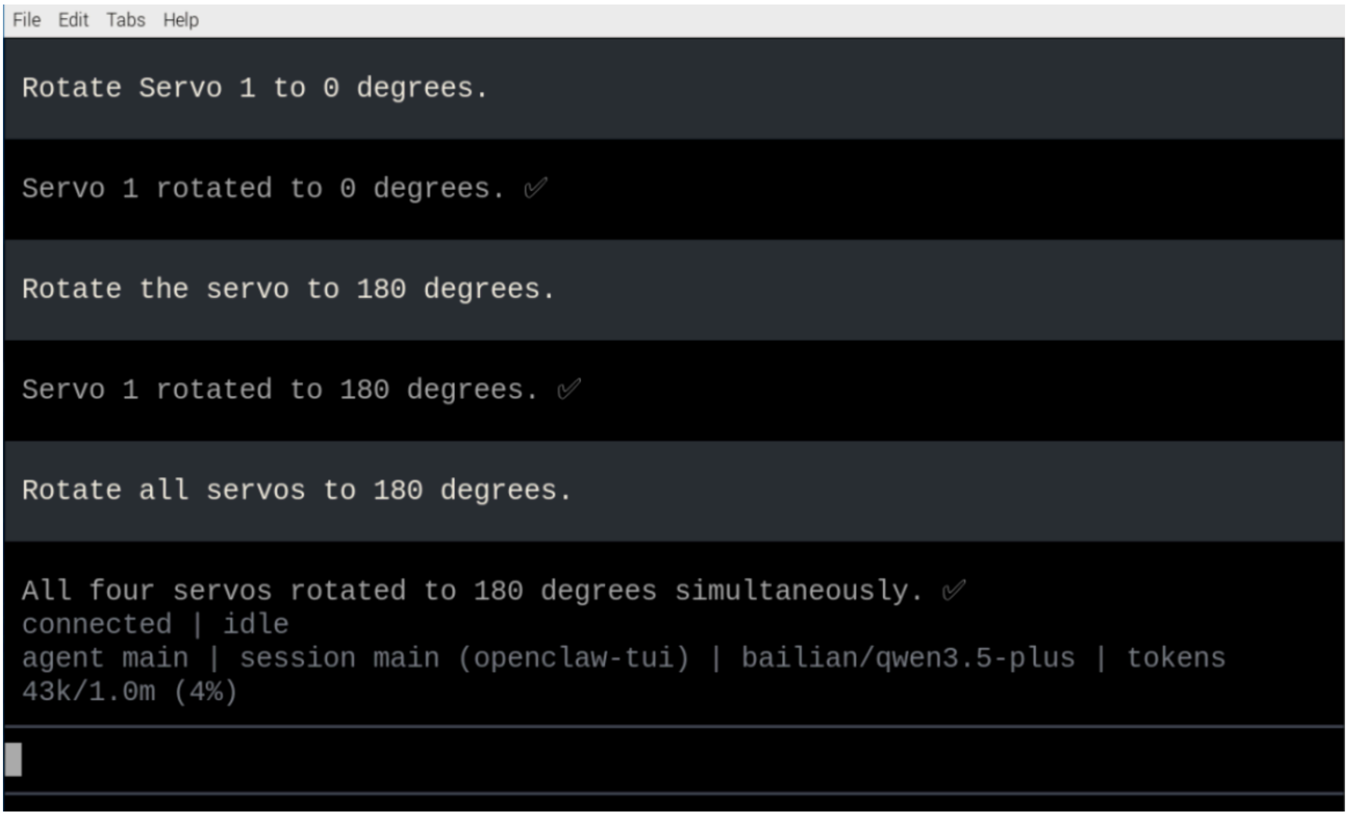
```
openclaw tui
```



After ensuring the hardware is connected, you can test servo control. Enter the following content in the terminal:

- Turn servo 1 to 0 degrees
- Turn servo to 90 degrees
- Turn servo to 180 degrees

Note: If possible, specify which servo number to control



If you can see the servo drive normally, it indicates the following are all normal.

- OpenClaw can receive commands
- Servo power supply is normal
- Servo angle control range is correct
- Clear feedback can be seen after command execution

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote control, classroom demonstrations	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the control chain
Voice code	Voice interaction, complete AI experience	Closest to real conversation experience	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

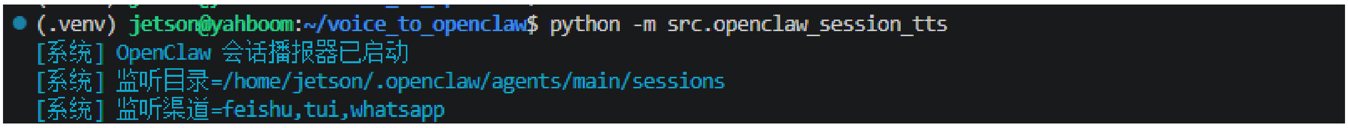
- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

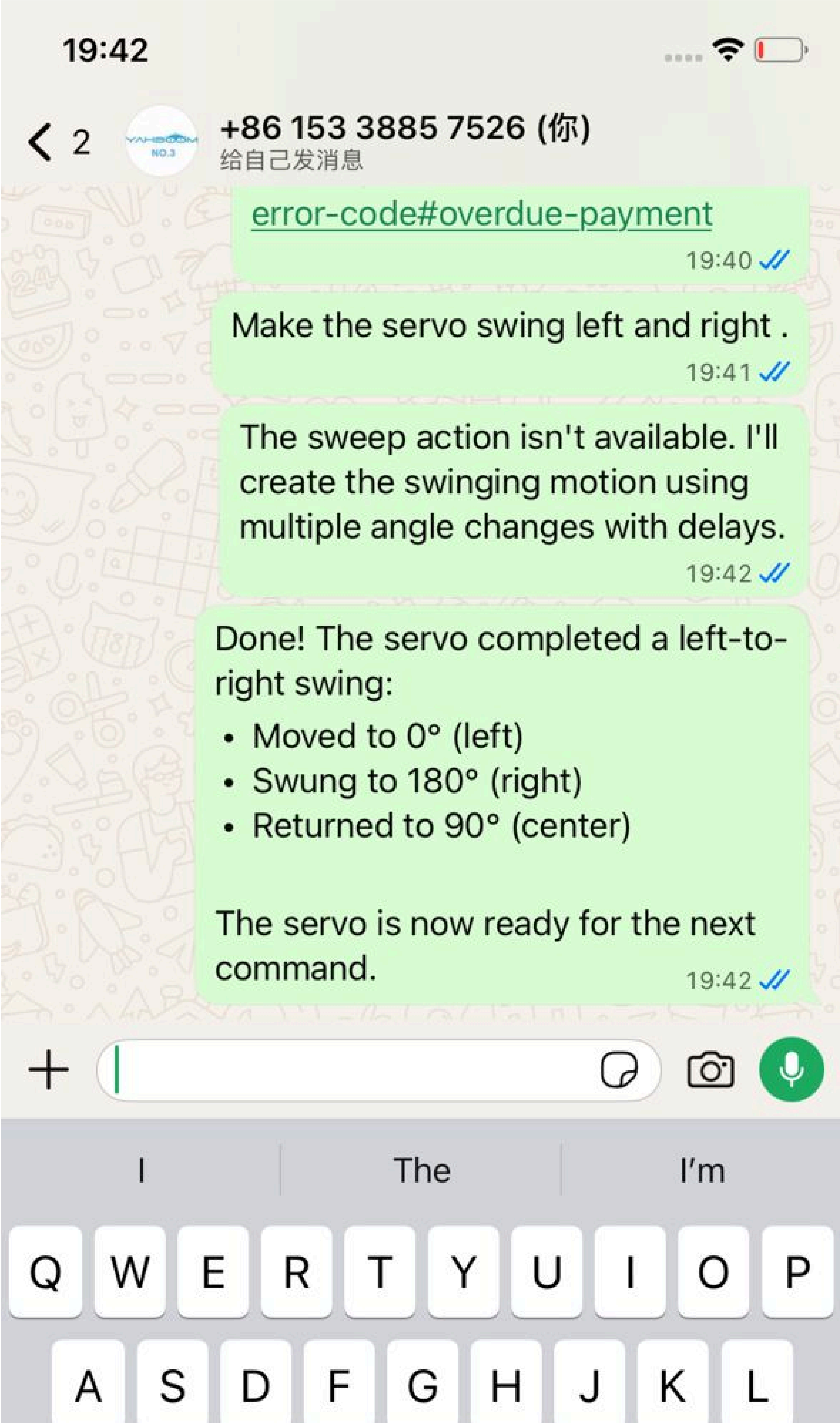


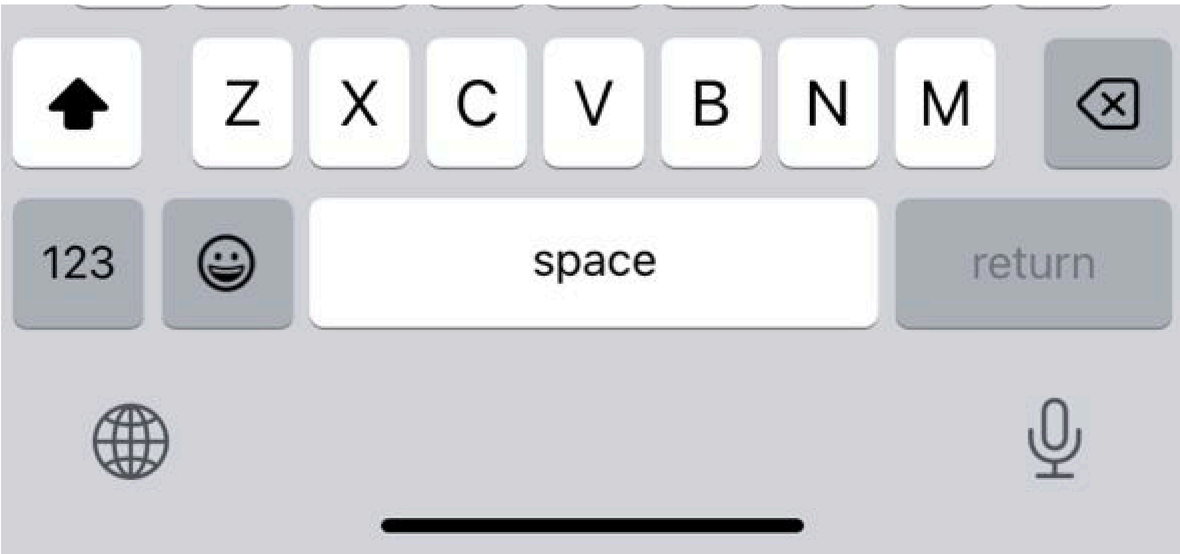
5.1 whatsapp Solution

For whatsapp connection, QR code creation, and gateway restart steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for the servo tutorial.

You can directly dialogue, for example:

- Turn servo 1 to 0 degrees
- Turn servo to 90 degrees
- Make the servo swing left and right

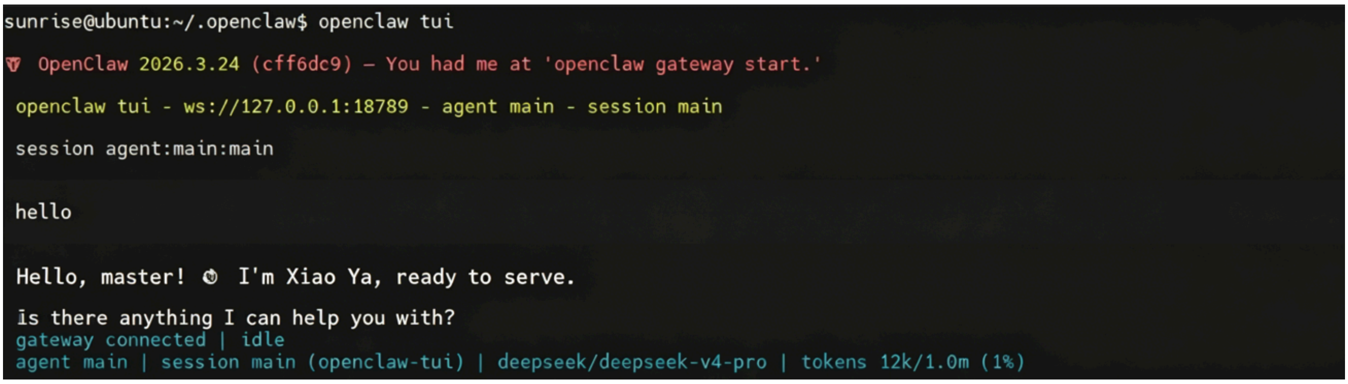

19:42
+86 153 3885 7526 (你)
给自己发消息
error-code#overdue-payment
19:40
Make the servo swing left and right .
19:41
The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.
19:42
Done! The servo completed a left-to-right swing:
• Moved to 0° (left)
• Swung to 180° (right)
• Returned to 90° (center)
The servo is now ready for the next command.
19:42
I
The
I'm
Q W E R T Y U I O P
A S D F G H J K L



5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter the following command in the terminal to continue the dialogue:

```
openclaw tui
```



5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. This is more suitable for directly demonstrating servo-type verbal commands, for example:

- Turn servo to 90 degrees
- Let the servo nod
- Make the servo swing left and right

```
2026-05-14 14:48:20 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw startin
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串
口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6060 ms 语音, 正在转写

[你] Make the servo swing left and right.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me make the servo swing using individual angle commands:

Done!
           The servo swung left to right and back to center.

The motion
           sequence was:
- Center (90°) → Left (0°) → Right (180°) → Center
           (90°)
[信息] 回复已裁剪为播报文本: 202 -> 116 字
```

6. Comprehensive Cases

Note: The following are all dialogue demonstrations using openclaw tui method (you can choose your own dialogue method)

6.1 Case One: Servo Dancing

Experimental Objective

Let the servo move continuously according to a set of preset angles, creating a rhythm effect similar to "dancing".

Effect Description

The servo can switch between multiple angles, such as swinging left, returning to center, swinging right, fast trembling, then returning to initial position. Although there is only one servo, through changes in action rhythm, it can still show an obvious "dance feel".

Example Instructions

- Let the servo dance for 10 seconds
- Let the servo swing left and right, faster rhythm
- Give the servo a three-beat dance step

6.2 Case Two: Yes/No Question Answering

Experimental Objective

Let the servo use actions to express "yes" or "no". For example, nodding represents "yes", shaking head represents "no".

Effect Description

When OpenClaw determines the answer is affirmative, the servo performs a nodding action; when the answer is negative, the servo performs a head-shaking action. This case is very suitable for highlighting the combination of "natural language understanding + action expression".

Example Instructions

- If the answer is "yes", let the servo nod
- If the answer is "no", let the servo shake its head
- Help me judge this question and answer with servo actions

6.3 Case Three: Table Tennis Simulation

Experimental Objective

Let the servo continuously perform left-right reciprocating motion, simulating the feeling of a table tennis ball bouncing back and forth on the table.

Effect Description

The servo swings uniformly or at variable speed between left and right boundary angles. If combined with voice description or screen prompts, the audience will easily understand this is a "table tennis ball moving back and forth" simulation.

Example Instructions

- Let the servo simulate table tennis ball moving back and forth
- Swing left and right 20 times, faster speed
- Start from the left, make a reciprocating motion demonstration

6.4 Case Four: Pointer Simulation

Experimental Objective

Use the servo as a rotatable pointer to indicate values, directions, or states.

Effect Description

This case can very intuitively demonstrate the mapping relationship of "abstract values -> actual angles". For example, mapping values from 0 to 100 to 0 to 180 degrees, the servo can point to the target position like a dashboard pointer.

Example Instructions

- Turn the pointer to 30%
- Let the servo point to the 90-degree position
- Simulate a dashboard, current value is 75